

Trabalho de CDI II - Séries

Exercício 1

a-) $\sum_{k=1}^{\infty} \frac{\ln k}{k} \rightarrow$ pelo critério da integral

$$f(x) = \frac{\ln x}{x}$$

$$\int_1^{\infty} \frac{\ln x}{x} dx = \lim_{a \rightarrow +\infty} \int_1^a \frac{\ln x}{x} dx = \lim_{a \rightarrow +\infty} \left(\frac{(\ln x)^2}{2} \right) \Big|_1^a = \lim_{a \rightarrow +\infty} \left(\frac{(\ln a)^2}{2} - \frac{(\ln 1)^2}{2} \right) = \infty - 0 = \infty$$

integral indefinida

$$\int \frac{\ln x}{x} dx = \int \frac{u}{x} x du = \int u du = \frac{u^2}{2} = \frac{(\ln x)^2}{2}$$

$$\ln x = u$$

$$du = \frac{1}{x} dx$$

pele critério da integral a
série diverge

$$du = dx$$

b-) $\sum_{k=1}^{\infty} \frac{k!}{k^k} \rightarrow$ pelo critério da razão $L = \lim_{k \rightarrow \infty} \frac{a_{k+1}}{a_k}$

$$L = \lim_{k \rightarrow \infty} \frac{(k+1)!}{(k+1)^{k+1}} \cdot \frac{k^k}{k!} = \lim_{k \rightarrow \infty} \frac{k^k (k+1)}{(k+1)^{k+1}} = \lim_{k \rightarrow \infty} \frac{k^k}{(k+1)^k} =$$

$$L = \lim_{k \rightarrow \infty} \frac{k^k}{(k+1)^k} = \lim_{k \rightarrow \infty} \left(\frac{k}{k+1} \right)^k = e^{-1} = \frac{1}{e} < 1$$

↳ propriedade de

$$\lim_{x \rightarrow \infty} \left(\left(\frac{x}{x+k} \right)^x \right) = e^{-k}$$

$$L < 1$$

↳ pelo teste da razão
converge

Exercício 2

$$a) \sum_{k=0}^{\infty} 2 \frac{3^k}{5^k} + 4 \frac{4^k}{10^k}$$

$$\sum_{k=0}^{\infty} 2 \frac{3^k}{5^k} + 4 \frac{4^k}{10^k} = \sum_{k=0}^{\infty} 2 \frac{3^k}{5^k} + \sum_{k=0}^{\infty} 4 \frac{4^k}{10^k} = \sum_{k=0}^{\infty} 2 \left(\frac{3}{5} \right)^k + \sum_{k=0}^{\infty} 4 \left(\frac{4}{10} \right)^k$$

$\underbrace{\hspace{10em}}_{\text{séries geométricas}}$

$\underbrace{\hspace{10em}}_{\substack{1 \\ q = \frac{3}{5} < 1}} + \underbrace{\hspace{10em}}_{\substack{2 \\ q = \frac{4}{10} < 1}}$

Domínio da série geométrica $S_n = a \cdot \frac{1 - q^{n+1}}{1 - q}$

converge

$$S_{n1} = 2 \cdot \frac{1 - \left(\frac{3}{5}\right)^{n+1}}{1 - \frac{3}{5}} = 2 \cdot \frac{1 - \frac{3}{5} \cdot \frac{3}{5} \cdot \frac{3}{5} \dots}{\frac{2}{5}} = 2 \cdot \frac{5}{2} = 5$$

$$S_n = S_1 + S_2 = 5 + 35 = 40$$

$$S_{n2} = 4 \cdot \frac{1 - \left(\frac{4}{10}\right)^{n+1}}{1 - \frac{4}{10}} = 4 \cdot \frac{1 - \frac{4}{10} \cdot \frac{4}{10} \cdot \frac{4}{10} \dots}{\frac{3}{5}} = 4 \cdot \frac{5}{3} = \frac{20}{3}$$

$$= 15 + \frac{20}{3} = \frac{65}{3}$$

b) $\sum_{k=0}^{\infty} \frac{(-5)^k}{(2k)!} \rightarrow$ pelo critério da razão

$$L = \lim_{k \rightarrow \infty} \frac{a_{k+1}}{a_k} = \lim_{k \rightarrow \infty} \frac{((-5)^{k+1} \cdot 2k!)}{(2(k+1))! \cdot (-5)^k} = \lim_{k \rightarrow \infty} \frac{(-5) \cdot 2k!}{(2(k+1))!} =$$

$$L = \lim_{k \rightarrow \infty} \frac{(-5) \cdot 2k!}{(2k+1)(2k+2)} = \lim_{k \rightarrow \infty} \frac{(-5)}{(2k+1)(2k+2)} =$$

$$L = \frac{-5}{(2 \cdot \infty + 1)(2 \cdot \infty + 2)} = \frac{-5}{\infty \cdot \infty} = \frac{-5}{\infty} = 0 < 1$$

↳ pelo critério da razão a série converge $\frac{n!}{(n+m)!} = \frac{1}{(n+1)(n+2)}$

Somar os primeiros 4 termos

$$S = \frac{(-5)^0}{(2 \cdot 0)!} + \frac{(-5)^1}{(2 \cdot 1)!} + \frac{(-5)^2}{(2 \cdot 2)!} + \frac{(-5)^3}{(2 \cdot 3)!}$$

$$= \frac{1}{0!} - \frac{5}{2!} + \frac{25}{4!} - \frac{125}{6!}$$

$$= 1 - \frac{5}{2} + \frac{25}{24} - \frac{125}{720}$$

$$= 1 - 2,5 + 1,0416 - 0,1736$$

$$= -2,0416 + 2,6436$$

$$= 0,602$$

Exercício 3 $\sum_{k=1}^{\infty} \frac{(x-3)^k}{k^2 5^k}$

$$C=3 \quad a_k = \frac{1}{k^2 5^k} \quad R = \lim_{k \rightarrow \infty} \left| \frac{a_k}{a_{k+1}} \right|$$

$$R = \lim_{k \rightarrow \infty} \left| \frac{1}{k^2 5^k} \cdot \frac{(k+1)^2 \cdot 5^{k+1}}{k^2 5^k} \right| = \lim_{k \rightarrow \infty} \left| \frac{(k+1)^2 \cdot 5^{k+1}}{k^2 5^k} \right| = \lim_{k \rightarrow \infty} \left| \frac{5(k+1)^2}{k^2} \right| =$$

$$R = \lim_{k \rightarrow \infty} \frac{5(k+1)^2}{k^2} = 5 \lim_{k \rightarrow \infty} \frac{(k+1)^2}{k^2} = 5 \lim_{k \rightarrow \infty} \left(k \left(1 + \frac{1}{k} \right) \right)^2 = 5 \lim_{k \rightarrow \infty} k^2 \left(1 + \frac{1}{k} \right)^2 =$$

$$R = 5 \lim_{k \rightarrow \infty} \left(1 + \frac{1}{k} \right)^2 = 5 \left(\lim_{k \rightarrow \infty} 1 + \lim_{k \rightarrow \infty} \frac{1}{k} \right)^2 = 5 \cdot 1^2 = 5$$

$$I^\circ =]C-R; C+R[=]3-5; 3+5[=]-2, 8[$$

$x = -2$ → série alternada-decrescente + teste da série alternada

$$\sum_{k=1}^{\infty} \frac{(-5)^k}{k^2 5^k} = \sum_{k=1}^{\infty} \frac{(-1)^k (5)^k}{k^2 5^k} \quad \lim_{k \rightarrow \infty} \frac{5^k}{k^2 5^k} = \lim_{k \rightarrow \infty} \frac{1}{k^2} = 0 \quad \left. \vphantom{\sum_{k=1}^{\infty}} \right\} \text{converge}$$

$$x = 8$$

$$\sum_{k=1}^{\infty} \frac{5^k}{k^2 5^k} = \sum_{k=1}^{\infty} \frac{1}{k^2} \quad p > 1 \quad \text{converge}$$

então

$$I^\circ = [-2, 8]$$

Exercício 4

$$a-) f(x) = \frac{x}{4+x^2} \quad -2 < x < 2$$

$$\frac{x}{4+x^2} = x \cdot \frac{1}{4+x^2} = x \cdot \frac{1}{4-(-x^2)} = x \cdot \frac{1}{4(1-(\frac{-x^2}{4}))} = \frac{x}{4} \cdot \frac{1}{1-(\frac{-x^2}{4})} =$$

$$\frac{x}{4} \cdot \frac{1}{1-u} \quad u = -\frac{x^2}{4}$$

$$\frac{1}{1-u} = 1 + u + u^2 + u^3 + u^4 + \dots$$

$$\frac{1}{1-(\frac{-x^2}{4})} = 1 - \frac{x^2}{4} + \left(\frac{-x^2}{4}\right)^2 + \left(\frac{-x^2}{4}\right)^3 + \left(\frac{-x^2}{4}\right)^4 + \dots$$

$$= 1 - \frac{x^2}{4} + \frac{x^4}{16} - \frac{x^6}{64} + \frac{x^8}{256} + \dots$$

$$\frac{x}{4} \cdot \frac{1}{1-(\frac{-x^2}{4})} = \frac{x}{4} \left(1 - \frac{x^2}{4} + \frac{x^4}{16} - \frac{x^6}{64} + \frac{x^8}{256} + \dots \right)$$

$$= \frac{x}{4} - \frac{x^3}{16} + \frac{x^5}{64} - \frac{x^7}{256} + \frac{x^9}{1024} + \dots$$

$$b-) \int \frac{\cos x - 1}{x} dx \quad \cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \frac{x^8}{8!} + \dots$$

$$\cos x - 1 = -\frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \frac{x^8}{8!} + \dots \quad \frac{\cos x - 1}{x} = -\frac{x}{2!} + \frac{x^3}{4!} - \frac{x^5}{6!} + \frac{x^7}{8!} + \dots$$

$$\int \frac{\cos x - 1}{x} dx = \int -\frac{x}{2!} + \frac{x^3}{4!} - \frac{x^5}{6!} + \frac{x^7}{8!} + \dots dx =$$

$$= -\frac{x^2}{2 \cdot 2!} + \frac{x^4}{4 \cdot 4!} - \frac{x^6}{6 \cdot 6!} + \frac{x^8}{8 \cdot 8!} + \dots + C$$